# The Emissions–Temperature Relationship

> Part of the Functional Series. Warming tracks the $CO_2$ humans have put into the air over time. This document explains how the platform measures that relationship from observed data, how it compares with the IPCC's own figure, and how much weight the number can carry. The input series are described in Concentrations and Temperature.

## 1. Cumulative, not annual

A single year's emissions say little about that year's temperature: $CO_2$ stays in the atmosphere for a very long time, so today's warming reflects everything emitted so far. The climate-science term for the resulting relationship is the **transient climate response to cumulative $CO_2$ emissions (TCRE)**: roughly, how many degrees of warming follow from each additional trillion tonnes of $CO_2$.

The platform measures a simplified, data-driven analogue of it: a straight-line fit of the observed temperature anomaly against the observed cumulative $CO_2$ emitted since 1850.

| | |
|---|---|
| **Temperature** (y) | Berkeley Earth global anomaly relative to 1850–1900, annual, 1850–2024 |
| **Emissions** (x) | World cumulative $CO_2$ since 1850: fossil fuels and cement, **plus land-use change**, from OWID/Global Carbon Project, including international aviation and shipping |
| **Fit** | Ordinary least squares with an intercept; 175 annual points |
| **Unit of the answer** | °C per 1,000 $GtCO_2$ (a thousand billion tonnes) |

## 2. The result

| Quantity | Value |
|---|---|
| Slope, total anthropogenic $CO_2$ | **0.486 °C per 1,000 $GtCO_2$** |
| 95% interval (robust to autocorrelation) | 0.442 – 0.530 |
| $R^2$ | 0.89 |
| Cumulative $CO_2$ since 1850, to 2024 | about 2,750 $GtCO_2$ total; 1,849 $GtCO_2$ fossil and cement only |
| Same fit, fossil and cement only | 0.749 (0.705 – 0.793) |

**Reading $R^2$ = 0.89.** Cumulative emissions account for most of the long-run variation in the temperature record. That is a statement of association over 175 years in which both series trend upward; it is not, by itself, proof of cause. The cause is established by physics and other evidence, and the platform states plainly that this is "context, not proof".

## 3. Comparison with the IPCC range

The IPCC's Sixth Assessment Report gives a TCRE best estimate of about 0.45 °C per 1,000 $GtCO_2$, with a very likely range of 0.27–0.63 (as quoted in the platform; the Benchmarks section, in development, re-checks it against the IPCC publication).

| Measure | Platform | Versus IPCC AR6 |
|---|---|---|
| Total-$CO_2$ slope | 0.486 | Inside the range; 1.08 × the best estimate; the 95% interval overlaps the range |
| Fossil-and-cement-only slope | 0.749 | Above the range |

The fossil-only figure sits higher because it leaves out land-use emissions: the same warming is being attributed to a smaller amount of $CO_2$, so each tonne appears to do more. The comparison is only meaningful for the total-$CO_2$ version, which matches the IPCC definition more closely, and even that is an analogue, not a restatement, because:

- the regression also absorbs warming from **non-$CO_2$ gases and aerosols** that rose and fell alongside $CO_2$;
- it uses **one observed climate history**, not an ensemble of Earth-system models;
- it depends on **land-use emission estimates** that are themselves uncertain (about ±2.6 $GtCO_2$ per year, one standard deviation, in recent decades).

## 4. How the uncertainty is handled

The 175 annual points are not independent: a warm year tends to follow a warm year. Ignoring that makes the fit look more certain than it is.

| Check | What it shows |
|---|---|
| Residuals are autocorrelated | Lag-1 autocorrelation 0.59; Durbin–Watson 0.80 (a value near 2 would mean none) |
| Robust standard error | Newey–West (HAC) standard error 0.022, against 0.013 from the naive formula: about 1.7 × larger, so the naive interval would be too narrow |
| Lag-length sensitivity | The interval is 0.447–0.524, 0.442–0.530 and 0.436–0.536 for 4, 8 and 16 lags |
| Block bootstrap | Resampling the residuals in 10-year blocks, 2,000 times, gives 0.435–0.536, close to the robust interval; other block lengths from 5 to 30 years give intervals of similar width |
| Holdouts | The fit is also tested by fitting on early decades and predicting later ones, against a simple baseline |

## 5. The relationship is not constant

Fitting only more recent periods gives larger slopes:

| Fit period | Slope (°C per 1,000 $GtCO_2$) |
|---|---|
| 1850–2024 | 0.486 |

| Fit period | Slope (°C per 1,000 $GtCO_2$) |
|---|---|
| 1900–2024 | 0.527 |
| 1950–2024 | 0.581 |
| 1970–2024 | 0.630 |

If $CO_2$ alone set the temperature through a fixed proportionality, the slope would not drift this way. The drift is consistent with other influences changing over time (non-$CO_2$ gases, aerosols, the pace of ocean heat uptake), and it means the headline number is the best single summary of the full record, not a law. The platform uses the full-range value for its scenario translation and shows the shorter windows so a reader can see the sensitivity.

## 6. What this is not

- Not a climate model: no physics, no ocean, no feedbacks.
- Not an attribution of warming to any country. Country shares describe each country's part of *cumulative emissions*; no country series is regressed against temperature.
- Not a forecast on its own. Its use in the scenario translation is described in Scenarios.

## 7. Where this appears in the dashboard

| Dashboard element | What it shows |
|---|---|
| Climate Correlation page | The relationship chart, the headline slope, and an IPCC AR6 comparison strip |
| "How this number was derived" | A seven-step explanation: what is regressed on what, the uncertainty, sensitivity, stability, the IPCC comparison, data and licences, and what the number is *not* |
| "Methodology & sources" | The sources table (coverage, vintage, licence), the baseline used by each view, the temperature offset and the two-totals note |

## See also

- Scenarios and Temperature Mapping
- Limitations and Uncertainty
- Technical: Multi-Source Data Pipelines: how the fit and its uncertainty are computed